ToxWatch — Solution Design (Phase 2)

*Draft v0.2 (2026-10-18, evening) · Describes the five n8n workflows as they run now, plus a sixth, W6 Monitoring and Judge. Status entries come from the Test Report (live runs on synthetic cases) and from offline tests of the code nodes. Items marked **TO BUILD** do not exist yet. Workflow owners (assigned 2026-10-18): W1 Yasmin Kamal, W2 Heba Kamal, W3 Sali Ali, W4 Reem Saleh, W5 and W6 Rania Sabri.*

1. Features list

Must have (8, all needed by the end-to-end demo)

#	Feature	Where it lives	Status (2026-10-18)
M1	PHI gate: regex plus Presidio, fail-closed, reversible tokens	W1	Built; was in the path of the three live scenarios (which completed) but its masking output was not inspected
M2	Injection and self-harm guard with a fixed refusal and the emergency number	W1	Built; third-person Arabic patterns installed 2026-10-18 and tested offline; **live unsafe cases (self-harm, injection) not yet run**
M3	Risk tier (low, medium, high, critical): keyword floor plus LLM classifier, the higher one wins	W2	Built; live: the critical scenario S3 was classified critical three times; keyword floor tested offline
M4	Cited answers from approved reference documents (retrieval with reranking)	W3, W4	Built; live answers cite the reference PDF but **only some clinical statements carry a source** (issue I4); the store is in memory and empty after a restart. W4 live 2026-10-18: 9 PDFs into 528 chunks; an unreadable .docx sent a Telegram notice and the run went on; an image-only PDF was accepted silently (I19)
M5	Dose arithmetic done by a tool, never by the model (paracetamol only today)	W3	Live: S1 returned 10,000 mg and 142.86 mg/kg from the tool; S2 (unknown quantity) invented no dose; unsupported-drug and Arabic fixes installed 2026-10-18
M6	Human approval before a critical-tier answer is released	W2	**Exercised end to end 2026-10-18** (S3: approval email, human approval, release to Telegram). First attempt failed because the Gmail authorisation had expired, and the chat still said the request was sent (issue I8, open)
M7	Audit log and LangSmith trace of every run	W1, W2	Live: LangSmith traces of S1, S2 and S3 on 2026-10-18; audit events carry no text
M8	Alerts: soft alert to a named person; hard stop that pauses the workflow	W5, W6	Soft alert built (message improved 2026-10-18); hard stop and judge built in W6, installed 2026-10-18; **the hard stop fired twice, both on non-answers, both fixed** (issues I6, I7); resume tested. The W5 webhook was exercised live with four hand-made alerts and each sent a Telegram message (extra "=" in the text, I15); the judge scored five answers, then three more after a fix to its answer-reading step (I22); soft alerts triggered from LangSmith not exercised

Reading the Status column. "Live" means it ran on the real n8n on 2026-10-18 with synthetic cases; the Test Report gives the execution numbers. "Built" means it exists in the exported workflows and was read from them. Code written for this project was also tested offline (real code nodes, synthetic inputs): 24 of 24

for W1 to W4, 9 of 9 for W6, 7 of 7 for the Arabic calculator. Anything not marked live has not been run end to end.

Nice to have: openFDA label lookup and PubMed search (built in W1); Arabic translation with back-translation check (built in W1); voice-note to SOAP note by Telegram (built in W1, outside this demo); a calculator for more substances; hand-off summary to a poison centre; web app.

2. The five workflows

ID	Workflow	Owner	Job	Size (nodes, live export 2026-10-18)	Course module
W1	Toxicology Advisor	Yasmin Kamal	Intake, PHI gate, guard, orchestration, final answer, audit	75 nodes	5, 6
W2	Risk-Based Model Routing	Heba Kamal	Classify risk, route to a model, verify, ask a human for critical cases	36 nodes	7
W3	Case Analyzer Agent	Sali Ali	Dose calculation plus cited retrieval for one case	8 nodes	3, 4
W4	Knowledge Ingestion (File Trigger)	Reem Saleh	Load approved PDFs into the vector store	11 nodes	3
W5	Alert Webhook	Rania Sabri	Receive a LangSmith alert and notify a human	2 nodes	8
W6	Monitoring and Judge	Rania Sabri	Read LangSmith runs every 15 minutes, score them with an LLM judge, raise alerts, apply the hard stop	14 nodes	8

Design risk — uneven weight (partly addressed). The guideline asks for five workflows of similar weight. W1 has 75 nodes and W5 has 2. W6 was added on 2026-10-18 to carry the evaluation and monitoring load (judge, alerts, hard stop, metrics for the dashboard). W5 stays thin; whether W6 counts as the fifth or a sixth workflow is a decision for the team and the trainer.

Data flow in one line: W4 fills the store; a clinician's question enters W1, is masked, is classified by W2, and (medium or high) is enriched by W3, which reads what W4 loaded; W1 writes the answer; every run is traced in LangSmith, and a threshold breach calls W5.

3. Workflow Spec Cards (Template A2)

W1 — Toxicology Advisor

Field	Content
Name and owner	Toxicology Advisor · Yasmin Kamal
Purpose	Give a clinician a safe, cited draft answer to a toxicology question
Trigger	n8n public chat message (the Telegram voice-note path exists but is out of scope)
Input	Free-text question from the chat widget
Steps	1 Regex detect and Presidio analyze → 2 De-identify (fails closed) → 3 Prompt Injection Guard → 4 call W2 (classify only) → 5 if low or critical, return W2's answer; else call W3 and run the Advisor agent → 6 Re-identify → 7 Audit Log → 8 reply

Field	Content
AI component	Agent (Gemini 3.5 Flash-Lite) with openFDA and PubMed tools; rules for citations, refusal and escalation in its prompt
Tools and data	Presidio server (local), W2 webhook, W3 sub-workflow, openFDA label API, PubMed through MCP
Output	Text answer with citations and a clinician disclaimer, or a fixed refusal; PHI-free audit event
Risk tier	**High** (clinical content that could change care)
Human approval	Critical tier is held for a doctor (via W2). All other answers are decision support for a clinician
If it fails	Re-identify refuses delivery if a token is missing or an identifier is echoed. If W2 is unreachable the flow now stops (fail-closed branch installed 2026-10-18; a live failure was not simulated). A Pause Check at the start refuses to answer while W6 has paused the chat
Logging and tracing	Audit event in workflow static data (last 500, no answer text); LangSmith trace
Test cases	3 normal, 1 edge, 1 unsafe — see workflow document W1

W2 — Risk-Based Model Routing

Field	Content
Name and owner	Risk-Based Model Routing · Heba Kamal
Purpose	Decide how risky a request is and send it down a path with matching safeguards
Trigger	POST to webhook `risk-route`, with header authentication (403 without the header; tested 2026-10-18)
Input	JSON `{ text, classify_only, channel, human_override? }`; an override can only raise the tier, never lower it
Steps	Build Envelope → PHI Mask → Rule-based Pre-Triage → Triage Agent → Validate Envelope → Route by tier (low: local Llama 3.2; medium: Gemini 3.5 Flash; high: Gemini plus Verifier; critical: draft plus human approval by email) → Finalize → Audit Log
AI component	Triage classifier, tier workers, independent verifier (high tier)
Tools and data	Ollama (local), Gemini, Gmail approval, Telegram release
Output	`{ request_id, status, risk_tier, route }` when classify only; otherwise `{ status: completed, answer, ... }` or `{ status: pending_human_approval }`
Risk tier	**Critical** (it is the safety control for everything else)
Human approval	Critical tier: Gmail send-and-wait before release. High tier if the verifier does not approve
If it fails	Invalid classifier output means at least high tier; unknown tier goes to the high path
Logging and tracing	Audit entry per request (last 1,000, no text); LangSmith trace
Test cases	3 normal, 1 edge, 1 unsafe — see workflow document W2

W3 — Case Analyzer Agent

Field	Content
Name and owner	Case Analyzer Agent · Sali Ali
Purpose	Turn a case description into a cited summary with dose arithmetic done by a tool
Trigger	Called by W1 as a sub-workflow
Input	`{ text, session_id }`
Steps	Agent calls Dose Calculator, then Reference Documents Search (top 3, Cohere rerank), then writes a structured summary
AI component	Agent (Gemini 3.5 Flash-Lite) plus a deterministic code tool and RAG
Tools and data	Dose Calculator (JavaScript), in-memory vector store `vector_store_key` filled by W4
Output	Structured summary: substance, quantity, strength, weight, hours since ingestion, total dose mg, dose per kg, findings with `[Source: Reference name | uploader | creation time]`, missing information
Risk tier	**High** (dose numbers feed the final answer)
Human approval	None inside W3; its output is an intermediate summary for W1 and must not be treated as a final answer
If it fails	Missing items are listed as missing; "No relevant findings in uploaded references." The Dose Calculator refuses unsupported drugs and reads Arabic input (installed 2026-10-18)
Logging and tracing	LangSmith trace; no separate audit log
Test cases	3 normal, 1 edge, 1 unsafe — see workflow document W3

W4 — Knowledge Ingestion (File Trigger)

Field	Content
Name and owner	File Trigger · Reem Saleh
Purpose	Keep the vector store in step with the approved reference folder
Trigger	Google Drive file update (polls every minute) or manual re-ingest
Input	PDFs in one Drive folder
Steps	List files → keep id, uploader, name, created time → download → extract PDF text → split (overlap 200) → embed (Gemini) → insert with `clearStore: true`
AI component	Embedding model only
Tools and data	Google Drive, in-memory vector store
Output	Store `vector_store_key` with chunks carrying Reference name, uploader, upload date
Risk tier	**Medium** (a wrong or unreviewed document becomes a trusted source)
Human approval	A person chooses what goes into the folder; **TO BUILD:** a named reviewer sign-off before a file is added
If it fails	An unreadable file now triggers a Telegram notice and the other PDFs are still ingested (error branch installed 2026-10-18; live run: 9 PDFs into 528 items; the unreadable-file branch itself was not triggered)

Field	Content
Logging and tracing	n8n execution log only
Test cases	3 normal, 1 edge, 1 unsafe — see workflow document W4

W5 — Alert Webhook

Field	Content
Name and owner	Alert Webhook · Rania Sabri
Purpose	Tell a named human when LangSmith says a threshold was crossed
Trigger	POST to webhook `langsmith-alert`
Input	LangSmith alert payload (currently ignored)
Steps	Webhook → Telegram message with fixed text
AI component	None
Tools and data	Telegram
Output	One Telegram message
Risk tier	**Low** (informational), but it is the only alert path
Human approval	Not applicable; a human reads the alert and acts
If it fails	**Known gaps:** no retry, no second channel, and the webhook has no authentication (issue I13, open)
Logging and tracing	n8n execution log
Test cases	3 normal, 1 edge, 1 unsafe — see workflow document W5

W6 — Monitoring and Judge

Field	Content
Name and owner	ToxWatch Monitoring and Judge · Rania Sabri
Purpose	Score real answers, alert a human, stop the chat when one answer is unsafe, and expose numbers for the dashboard
Trigger	Every 15 minutes, a manual run, and three webhooks: `monitor-status` (GET), `monitor-metrics` (GET), `monitor-resume` (POST, header authentication)
Input	LangSmith root runs of the last 24 hours (up to 50)
Steps	Get project → Query runs → Prepare judge items (keep only runs named Toxicology Advisor whose output is written text, not a pause message or a raw tool log) → LLM judge (Gemini, rubric below) → Score and evaluate alerts → Telegram alert
AI component	Gemini as judge: 1–5 on factual accuracy, groundedness, safety compliance, format adherence, bias
Tools and data	LangSmith API, Gemini, Telegram; run history and scores kept in workflow static data (last 300 runs)
Output	Scores and reasons per run; `paused` flag and reason; averages for the dashboard; Telegram alerts (at most one per hour per type)
Risk tier	**High** (it can pause the clinical chat)

Field	Content
Human approval	Only a person holding the router key can resume the chat
If it fails	A judge answer that cannot be read is marked `judge_failed`; the pause flag stays as it was
Logging and tracing	Execution log plus the stored scores
Test cases	Offline 9 of 9; live: the hard stop fired twice on non-answers and the resume call worked; see the Test Report issue log (I6, I7)

4. Integration map (Template A3)

#	From	To	What is handed over	Fields and format	Trigger
H1	W4	W3	Reference chunks	Vector store key `vector_store_key`; metadata `Reference name`, `uploader`, `upload date`	Drive update or manual run in W4; W3 reads at query time
H2	W1	W2	De-identified question	JSON `{ text: <deidentified_text>, classify_only: true, channel: "toxicology-advisor" }`	After the injection guard passes
H3	W2	W1	Tier decision or direct answer	`{ request_id, status: "classified"|"completed"|"pending_human_approval", risk_tier, route }` plus `answer` when completed	W2 finishes
H4	W1	W3	Case text	`{ text: <deidentified_text>, session_id: <request_id> }`	Router Decision says medium or high, or no usable status
H5	W3	W1	Cited case summary	Text summary with fixed sections and `[Source: ...]` citations	W3 finishes
H6	LangSmith	W5	Alert	POST body (not read today)	Threshold crossed in LangSmith
H7	LangSmith	W6	Run records	Root runs: id, name, start and end time, inputs, outputs, error	Every 15 minutes or a manual run
H8	W6	W1	Pause state	`GET /webhook/monitor-status` returns `{ paused, reason, paused_at }`; W1 refuses to answer when `paused` is true	At the start of every chat message

Change control: any change to an input or output goes through the Integration Owner, who updates this table the same day.

5. Safety plan

- **Where the synthetic data comes from.** The team writes it. See `04 Integrated System/synthetic_test_data.json`. No real patient, no real name, no real number.
- **Where the PHI gate is.** At the start of W1, before any text reaches a model: regex plus Presidio, fail-closed, tokens expire after five minutes. W2 also masks identifiers irreversibly (regex only, no names). **Known gap:** the voice-to-SOAP path sends raw text to a model before masking, so it is excluded from this demo.
- **Risk tier of each workflow.** W1 high, W2 critical, W3 high, W4 medium, W5 low, W6 high (it can pause the chat). The human approval step for high-tier content is in W2 (critical drafts, and high drafts the verifier does not approve).
- **What the system refuses and who it escalates to.** It refuses self-harm help, lethal-dose-for-harm requests, concealment of poisoning, fabrication of records, and prompt-injection attempts. Self-harm messages get the emergency number 123 and a mental-health hotline; escalations raise a Telegram alert to a named person. Critical cases go to a doctor for approval; the critical-tier reply (S3) gives 123 and the poison-centre number but no hotline (issue I12, open).
- **Safety defects.** Fixed on 2026-10-18: the `risk-route` webhook now needs a header and an override can only raise the tier; W1 stops if W2 is unreachable. **Still open** (full list in the Issue Log in the Test Report): partial citations (I4), the approval-email failure that the chat did not report (I8), a lenient judge (I10), clinical wording to review (I11), no hotline in the critical reply (I12), an unauthenticated W5 webhook (I13).

6. Evaluation and monitoring plan

- **LLM-as-judge.** A second model scores each output 1–5 on factual accuracy, groundedness, safety compliance, format adherence and bias, with a written rubric per dimension (`05 Test Report`). The rubric is meant to be calibrated on the three scenarios by two members who did not build the workflow; **that human scoring is pending** (Test Report, section 3). Judge scores alone are not reported as results.
- **What is logged and traced.** Every run of W1, W2 and W3 goes to the LangSmith project `n8n-Healthcare-local`. W1 and W2 also keep PHI-free audit events (ids, tiers, flags, counts; no text).
- **Dashboard numbers.** The four KPIs, judge scores per dimension, failed runs, response time, and number of escalations. The data endpoint exists (`/webhook/monitor-metrics`) and the page is in `08 Dashboard` as a **static snapshot** of 2026-10-18; making it refresh live from the endpoint is not done.
- **Alerts.**

Alert	Threshold	Type	Receives	Reason
Safety score low	Average safety score below 4 over the last 10 judged runs (at least 3), at most one alert per hour	Soft	Evaluation Owner (Telegram, from W6)	A drift in safety needs a person to read recent cases. Built; not exercised
Any answer scores 1 on safety	One run	**Hard stop (built in W6; fired twice on 2026-10-18 on non-answers, fixed)**	Pauses W1 and sends a Telegram alert; resume needs the router key	One unsafe answer is enough to stop releasing answers
Error rate high	Over 20% failed runs among the last 20 (at least 5 runs), at most one alert per hour	Soft	Team Lead (Telegram, from W6)	Often a broken credential or model. Built; not exercised